A Hybrid Machine Learning, Deep Learning, And Graph Neural Network Framework for HIV Drug Resistance Prediction and Mutation Interaction Modeling

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1. Introduction

HIV drug resistance remains a major clinical challenge due to the virus's high mutation rate, particularly in key enzymes such as protease, reverse transcriptase, and integrase. Accumulation of mutations in these proteins can lead to treatment failure. Computational prediction of resistance using machine learning (ML), deep learning (DL), and graph neural networks (GNN) is therefore essential, but requires biologically coherent and non-redundant datasets.

Beyond dataset curation, this work proposes a hybrid framework that models both sequence context and mutation interactions. Contextual embeddings are extracted using a pretrained protein transformer (ESM-2), while mutation interaction graphs are learned using GNNs. These representations are fused to perform drug-specific binary resistance prediction in a biologically realistic manner.

2. Dataset Used and Biological Relevance

2.1 Source

The dataset was taken from the Stanford HIV Drug Resistance Database (HIVDB), which provides curated genotype–phenotype relationships between HIV mutations and drug resistance which means it maps HIV mutations with how well the drug will work on them .

Each record includes:

- Sequence ID
- Patient ID
- Positional amino acid fields (P1–Pn)
- Mutation list (CompMutList)
- Drug susceptibility measurements

2.2 Drug Classes Included

Drug Class	Target Enzyme
PI	Protease
NRTI	Reverse Transcriptase

NNRTI	Reverse Transcriptase
INI	Integrase

These classes are biologically meaningful as each drug class targets a specific viral enzyme which has a unique evolutionary pattern and different resistance mechanisms.

2.3 Dataset

The dataset directly supports our research objective because:

- All sequences belong to HIV-1.
- Each dataset contains sequences from a single functional protein.
- Positional columns represent aligned amino acid residues.
- Drug resistance values are linked to each sequence.

Protein alignment lengths confirm biological specificity:

- Protease (PI): 99 amino acids
- Reverse Transcriptase (NRTI, NNRTI): 560 amino acids
- Integrase (INI): 288 amino acids

These match canonical HIV protein lengths, confirming no cross-protein contamination.

3. Methodology

3.1 Sequence Construction and Alignment Validation

Each dataset contained aligned positional columns (P1–Pn). These were combined to reconstruct full-length protein sequences.

Special handling was applied:

- Multi-residue entries (e.g., “LM”) were resolved by retaining the first residue.
- The symbol “-”, which represents wildtype (same as reference), was replaced using a consensus residue to ensure biologically valid sequence reconstruction.

Uniform sequence lengths were verified before further analysis.

3.2 Removal of Exact Duplicate Sequences

Sequences with identical amino acid patterns were removed.

This ensured:

- Each row represented a unique viral genotype.
- Repeated clinical isolates did not bias analysis.

3.3 Phenotype Filtering

Rows lacking drug-resistance values were excluded.

This ensured:

- Each retained sequence had associated phenotype data.
- The dataset remained suitable for supervised learning.

3.4 Identity-Based Redundancy Removal

Redundancy was defined using sequence identity:

Identity = (Number of matching amino acids) / (Total sequence length)

A greedy identity-based clustering approach was applied.

To avoid excessive data loss in highly conserved proteins, an adaptive identity threshold was used:

- Initial threshold: 95%
- Increased incrementally (up to 99%) if dataset size fell below 500
- Ensured minimum 500 sequences per drug class

3.5 Quantitative Identity Distribution Analysis

To validate redundancy removal, we computed:

- Mean pairwise identity before filtering
- Mean pairwise identity after filtering
- Percentage of high-identity pairs (≥ threshold) before and after filtering

This step ensured that redundancy removal was statistically validated rather than inferred only from dataset size reduction.

For example:

- INI dataset: ~98% of pairs ≥95% identity before filtering
- After adaptive filtering: 0% of pairs ≥99% identity

This confirms removal of near-identical clusters.

3.6 Feature Extraction and Representation Learning

This project models HIV drug resistance using two complementary feature types:

3.6.1 Contextual Sequence Embeddings (ESM-2)

To capture global protein context, full-length reconstructed sequences were passed through a pretrained protein transformer model, ESM-2 (esm2_t33_650M_UR50D).

BERT Model Used

ESM-2 is a protein language model trained on millions of sequences using a transformer architecture similar to BERT. It learns contextual amino acid representations capturing:

• Long-range dependencies
• Evolutionary constraints
• Biochemical patterns
• Structural tendencies

Available ESM-2 variants include:

Model	Parameters	Output Dimension
esm2_t6_8M	8M	320

esm2_t12_35M	35M	480
esm2_t30_150M	150M	640
esm2_t33_650M	650M	1280
esm2_t36_3B	3B	2560

We selected **esm2_t33_650M** because:

• It provides high representational capacity
• It balances accuracy and computational feasibility
• It is widely validated for protein representation tasks
• It produces a stable 1280-dimensional embedding

Extracted Feature

For each sequence:

ESM_Global_Embedding
(1280-dimensional vector)

This embedding was obtained by:

1. Feeding reconstructed protein sequence into ESM-2
2. Extracting final layer token representations
3. Applying mean pooling across residues

This feature captures the global biological context of the protein.

3.6.2 Explicit Mutation Features

To explicitly represent mutation identity, sequences were compared against canonical reference proteins.

Extracted features include:

• Mutation_Position_Index
• Reference_Amino_Acid
• Mutated_Amino_Acid
• Mutation_Type (e.g., K103N)
• Binary_Mutation_Vector (per alignment position)

These features ensure the model explicitly knows which resistance-associated mutations are present.

3.6.3 Mutation Interaction Graph Features

To model epistatic interactions, mutation graphs were constructed for each sequence.

Graph Design:

• Nodes = mutated positions
• Edges = co-occurring mutations within the same sequence

Node features include:

• Position index
• Mutated amino acid identity

A Graph Neural Network produces:

GNN_Mutation_Embedding
(interaction-aware representation)

This captures non-linear mutation interactions influencing resistance.

The ESM_Global_Embedding and GNN_Mutation_Embedding are concatenated to form a unified feature

vector. This fused representation is used for drug-specific binary classification.Separate binary classifiers are trained for each drug to account for drug-specific mutation-response relationships.

4. Results

4.1 Dataset Size

Drug Class	Initial Size	Final Size	Final Threshold
PI	4350	1216	95%
INI	1986	589	99%
NRTI	5355	552	97%
NNRTI	4911	538	97%

All final datasets contain more than 500 sequences, satisfying the assignment requirements.

4.2 Interpretation

Protease retained larger diversity, indicating higher mutation variability.

Integrase showed strong conservation, requiring stricter threshold adjustment.

Reverse transcriptase datasets showed substantial redundancy, consistent with clinical sampling patterns.

Overall, redundancy removal preserved mutation diversity while eliminating near-duplicate bias.

4.3 ESM-2 Embedding Extraction Results

Contextual embeddings were successfully extracted for all four drug classes.

Drug Class	Final Sequences	Embedding Dimension
PI	1216	1280
NRTI	552	1280
NNRTI	538	1280
INI	589	1280

No sequences were discarded due to reconstruction errors. Alignment lengths matched canonical protein lengths, confirming biologically valid input to the transformer model.

These embeddings serve as global sequence-level features for downstream hybrid modeling.

5. Code

The preprocessing pipeline includes:

- Sequence reconstruction from aligned positional columns
- Reference-aware handling of wildtype positions
- Exact duplicate removal

- Phenotype validation
- Adaptive greedy identity clustering
- Identity distribution analysis

6. Conclusion

This work produced biologically validated, non-redundant HIV drug-resistance datasets across four drug classes by removing exact duplicates,applying adaptive identity-based clustering, quantitatively validating identity distribution changes, confirming biological target specificity.

These cleaned datasets provide a strong foundation for hybrid deep learning and graph-based mutation interaction modeling for HIV drug resistance prediction.

In addition to dataset curation, contextual protein embeddings were extracted using a pretrained ESM-2 transformer model. These embeddings, combined with explicit mutation extraction and graph-based interaction modeling, provide the feature foundation for hybrid ML, DL, and GNN resistance prediction.